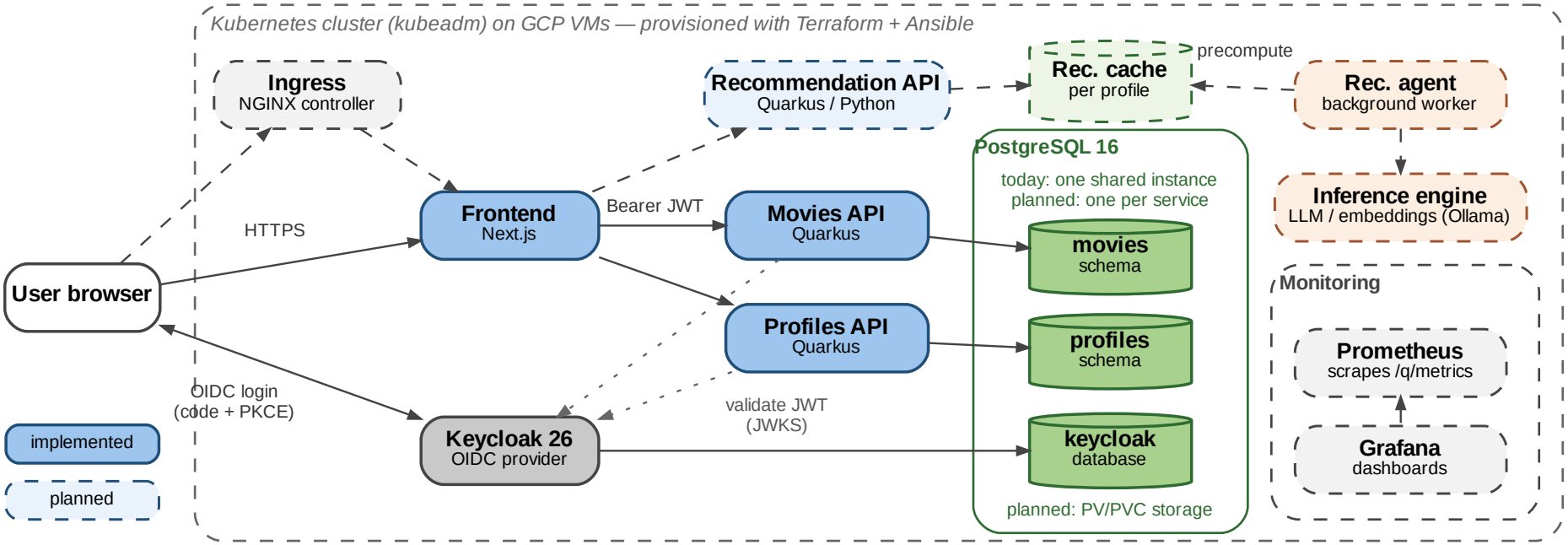
Kubernetes cluster (kubeadm) on GCP VMs — provisioned with Terraform + Ansible
Ingress
NGINX controller
Recommendation API
Quarkus / Python
Rec. cache
per profile
precompute
Rec. agent
background worker
PostgreSQL 16
today: one shared instance
planned: one per service
Inference engine
LLM / embeddings (Ollama)
HTTPS
Frontend
Next.js
Bearer JWT
Movies API
Quarkus
movies
schema
User browser
Profiles API
Quarkus
profiles
schema
Monitoring
Prometheus
scrapes /q/metrics
OIDC login
(code + PKCE)
validate JWT
(JWKS)
Keycloak 26
OIDC provider
keycloak
database
Grafana
dashboards
planned: PV/PVC storage
implemented
planned